# MILITARY SPECIFICATION

## MIXED GAS, DIVER'S BREATHING

### 1. SCOPE

1.1 Scope. This specification covers diver's breathing mixed gas.

1.2 Classification. Gas mixtures shall be of the following mixes, as specified (see 6.2):

Mix I - Helium, 68 percent; oxygen, 32 percent.
Mix II - Helium, 60 percent; oxygen, 40 percent.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-V-2 - Valve, Cylinder, Gas (for Compressed or Liquified Gases) General Specification for.
MIL-D-3716 - Desiccants, Activiated for Dynamic Dehumidification.
MIL-T-27730 - Tape, Antiseize, Tetrafluoroethylene, with Dispenser.

STANDARDS

MILITARY
MIL-STD-101 - Color Code for Pipelines and for Compressed Gas Cylinders.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

U. S. PHARMACOPOEIA CONVENTION PUBLICATION (USP)
U. S. Pharmacopoeia

(Application for copies should be addressed to the United States Pharmacopoeial Convention, 46 Park Avenue, New York, New York 10016.)

DEPARTMENT OF TRANSPORTATION (DOT)
49 CFR 171-190 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

### 3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, a first cylinder shall be tested as specified in 4.3 (see 6.5).

3.2 Purity.



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3.2.1 Mix I. Helium oxygen mix shall be 32 percent oxygen, balance helium moisture and permissible minor constituents, ± 0.5 percent of the oxygen component when tested in accordance with 4.6.4.

3.2.2 Mix II. Helium oxygen mix shall be 40 percent oxygen, balance helium moisture and permissible minor constituents, ± 1.2 percent of the oxygen component when tested in accordance with 4.6.4.

3.3 Impurities. The mixture shall be free from all contaminants of known toxicity to the maximum practicable extent. Limitations on specific impurities shall be as follows:

3.3.1 Odor.

3.3.1.1 Neither mixture shall contain an odor when tested in accordance with 4.6.2.

3.3.2 Minor constituents. The mixtures shall not contain minor constituents in excess of the quantities specified in table I.

Table I - Constituent concentrations.

| Constituent | Maximum concentration in ppm (by volume) |
|---|---|
| Nitrogen, argon, neon, hydrogen | 1970 |
| Carbon monoxide (CO) | .04 |
| Carbon dioxide ($CO_2$) | 4.8 |
| Methane ($CH_4$) | 20.6 |
| Acetylene ($C_2H_2$) | .04 |
| Ethylene ($C_2H_4$) | .16 |
| Ethane ($C_2H_6$) and other hydrocarbons expressed as ethane | 2.4 |
| Nitrous oxide | 0.8 |
| Halogenated compounds: | 0.8 |
| Refrigerants (Freons, etc.) | |
| Solvents (trichloroethylene, carbon tetrachloride, etc.) | 0.08 |
| Other (each discernible from background noise on infrared spectrophotometer) | 0.08 |
| Water | 6.25 |

3.3.3 Moisture. The moisture content shall not exceed 0.005 milligrams of water vapor per liter of gas at standard conditions (70°F., 1 atom) when tested as sepcified in 4.6.5.

3.3.4 Desiccants. When desiccants are used in contact with the mixtures, they shall meet the particle strength and corrosiveness requirements of type I, II, and IV desiccants specified in MIL-D-3716.

3.4 Containers.

3.4.1 Cylinders shall be furnished or obtained as specified (see 6.2), and shall be processed prior to filling as specified in 3.4.2. They shall be equipped with a CGA 350 outlet connection.

3.4.2 Processing cylinders.

3.4.2.1 Hydrostatic test date. Each cylinder shall be examined to determine the last date of hydrostatic testing. If the last date of testing is 5 or more years prior to the intended filling date, the cylinder shall be tested as specified in 3.4.2.2. If the last date of testing is less than 5 years prior to the intended filling date, the cylinder shall be processed as specified in 3.4.2.3.

3.4.2.2 Hydrostatic testing. When specified (see 6.2), cylinders that have not been hydrostatically tested within 5 years prior to the intended filling date shall be hydrostatically tested in accordance with the detailed procedure specified in the contract or order. Each cylinder subjected to the hydrostatic test shall be dried as specified in 3.4.2.6. If the contract or order does not specify hydrostatic testing, the cylinder requiring hydrostatic test in accordance with 3.4.2.1 shall be considered unsatisfactory for filling and shall be tagged as specified in 3.4.4.

3.4.2.3 Pressurization and odor inspection. Each cylinder shall be tested for the presence of odor as specified in 4.6.2 and inspected to determine the existence of a minimum internal pressure of 5 pounds per square inch gage (psig) or sufficient pressure to produce an audible hissing sound when the valve is opened. If the audible method is used, the palm of the hand or a slip of paper shall be used to insure that any hissing sound heard when the valve is opened is due to a positive pressure and not a vacuum. Cylinders having both sufficient internal pressure and no odor present in the gas may be inverted and the remainder of the internal



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pressure used to force any separated water out through the cylinder valve. Cylinders that do not have the required internal pressure or that have an odor or contain water, shall be dried as specified in 3.4.2.6. Cylinders having no odor and with required internal pressure and no indication of moisture shall be processed as specified in 3.4.2.4 and 3.4.2.7.

3.4.2.4 Cylinder valve inspection. Each cylinder valve shall be inspected for proper condition and functioning. If a valve is found to be damaged, inoperative, or otherwise unsatisfactory, a cylinder valve conforming to MIL-V-2 shall be installed, except where minor adjustment will put the unsatisfactory valve in satisfactory condition.

3.4.2.5 Cylinder valve installation. When installing a valve, antiseize tape conforming to MIL-T-27730 shall be used on the male threads. The tape shall be applied in accordance with application instruction of MIL-T-27730. The valve shall be tightened into the cylinder with a torque of 250 to 275 foot-pounds.

3.4.2.6 Cylinder drying. When specified herein, the cylinder shall be dried by either the hot gas purging method specified in 3.4.2.6.1 or the evacuation method specified in 3.4.2.6.2. If an odor is detected in a cylinder and the odor does not disappear in the drying process, the cylinder shall be considered unsatisfactory for filling and shall be tagged as specified in 3.4.4. Cylinders having no odor after being subjected to the drying process shall be inspected for painting and marking as specified in 3.4.2.7.

3.4.2.6.1 Hot gas purging method. The cylinder shall be dried by removing the cylinder valve, inserting a tube extending approximately to the bottom of the cylinder, and passing clean, hot, dry, oil-free air or nitrogen through the tube. The escaping air or nitrogen shall be smelled during the process to detect odor that could be present. At the end of the drying process, the cylinder valve shall be immediately replaced in accordance with 3.4.2.5 and closed.

3.4.2.6.2 Evacuation method. The cylinder shall be dried by attaching a vacuum pump to the cylinder valve outlet, opening the valve, and heating the cylinder to approximately 180°F. in an oven or hot water bath while evacuating the cylinder to an absolute pressure of approximately 50 psig pressure with oxygen, and the valve closed. The cylinder shall be permitted to stand for 30 minutes and then tested for odor in accordance with 4.6.2.

3.4.2.7 Repainting inspection. Each cylinder shall be inspected for proper painting and marking as required by MIL-STD-101 (see 5.2). Cylinders found to be unsatisfactory shall be painted and marked as required. Painting and marking may be performed either before or after filling. In either event, the cylinders shall be evacuated in accordance with 3.4.2.8 before filling.

3.4.2.8 Cylinder evacuation. Each cylinder shall be evacuated to an absolute pressure of less than 3 inches Hg and maintained in that condition until filling begins.

3.4.3 Inspection and cleaning of containers. The contractor shall perform his usual inspection and cleaning to assure that containers are free from contamination and suitable for shipment and storage. Containers with markings that indicate the container may have been used for other than diver's breathing mixed gas shall not be refilled and shall be evacuated as specified in 3.4.2.8.

3.4.4 Unsatisfactory containers. Government-owned containers found unsatisfactory for filling or refilling shall be tagged with a tag bearing an explanation of the reason for rejection. The tag shall be firmly wired to the cap of the cylinder.

3.5 Filling containers.

3.5.1 Mixed gas. Mixed gas cylinders that have been processed as specified in 3.4.3 shall be filled with mix I or mix II to a pressure of 1,800, 2,000, 2,200, or 2,400 psig as specified (see 6.2). When filling cylinders at a temperature other than 70°F., the fill pressure shall be as specified in table II for the applicable temperature and nominal fill pressure. When tested as specified in 4.6.3, the cylinder pressure shall not differ more than 125 psig from the pressure specified in table II that corresponds to the stabilized cylinder temperature.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Mixtures shall be subjected to the following:

(a) First article inspection (4.3)
(b) Quality conformance inspection (4.4)



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TABLE II - MIXED GAS CYLINDERS FILL PRESSURE FOR VARIOUS TEMPERATURES.

| MIX I | | | | | | MIX II | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Helium, 68 percent; oxygen, 32 ± 0.5 percent | | | | | | Helium, 60 percent; oxygen, 40 ± 1.2 percent | | | | | |
| Stabilized Temperature °F | Fill Pressure (psig) | | | | | Stabilized Temperature °F | Fill Pressure (psig) | | | | |
| | 1800 | 2000 | 2200 | 2400 | 2640 | | 1800 | 2000 | 2200 | 2400 | 2640 |
| 30 | 1654.137 | 1837.229 | 2020.151 | 2202.879 | 2421.984 | 30 | 1656.839 | 1840.553 | 2023.96 | 2207.364 | 2427.444 |
| 32 | 1661.429 | 1845.36 | 2029.149 | 2212.745 | 2432.887 | 32 | 1663.983 | 1848.516 | 2032.771 | 2217.001 | 2438.039 |
| 34 | 1668.721 | 1853.49 | 2038.147 | 2222.611 | 2443.79 | 34 | 1671.126 | 1856.479 | 2041.581 | 2226.639 | 2448.673 |
| 36 | 1676.012 | 1861.621 | 2047.144 | 2232.478 | 2454.692 | 36 | 1678.27 | 1864.443 | 2050.392 | 2236.276 | 2459.288 |
| 38 | 1683.304 | 1869.751 | 2056.142 | 2242.344 | 2465.595 | 38 | 1685.413 | 1872.406 | 2059.202 | 2245.914 | 2469.902 |
| 40 | 1690.596 | 1877.882 | 2065.14 | 2252.21 | 2476.498 | 40 | 1692.557 | 1880.369 | 2068.013 | 2255.551 | 2480.517 |
| 42 | 1697.904 | 1886.151 | 2074.14 | 2262.08 | 2487.419 | 42 | 1699.743 | 1888.37 | 2076.827 | 2265.187 | 2491.182 |
| 44 | 1705.211 | 1894.42 | 2083.139 | 2271.95 | 2498.341 | 44 | 1706.929 | 1896.371 | 2085.641 | 2274.823 | 2501.847 |
| 46 | 1712.519 | 1902.69 | 2092.139 | 2281.819 | 2509.262 | 46 | 1714.115 | 1904.371 | 2094.454 | 2284.46 | 2512.513 |
| 48 | 1719.826 | 1910.959 | 2101.138 | 2291.689 | 2520.184 | 48 | 1721.301 | 1912.372 | 2103.268 | 2294.096 | 2523.178 |
| 50 | 1727.134 | 1919.228 | 2110.138 | 2301.559 | 2531.105 | 50 | 1728.487 | 1920.373 | 2112.082 | 2303.732 | 2533.843 |
| 52 | 1734.432 | 1927.265 | 2119.131 | 2311.421 | 2541.922 | 52 | 1735.644 | 1928.35 | 2120.885 | 2313.358 | 2544.442 |
| 54 | 1741.729 | 1935.302 | 2128.124 | 2321.282 | 2552.739 | 54 | 1742.801 | 1936.326 | 2129.686 | 2322.985 | 2555.041 |
| 56 | 1749.027 | 1943.339 | 2137.118 | 2331.144 | 2563.557 | 56 | 1749.958 | 1944.303 | 2138.492 | 2332.611 | 2565.641 |
| 58 | 1756.324 | 1951.376 | 2146.111 | 2341.005 | 2574.374 | 58 | 1757.115 | 1952.279 | 2147.295 | 2342.238 | 2576.24 |
| 60 | 1763.622 | 1959.413 | 2155.104 | 2350.867 | 2585.191 | 60 | 1764.272 | 1960.256 | 2156.098 | 2351.864 | 2586.839 |
| 62 | 1770.898 | 1967.53 | 2164.083 | 2360.694 | 2596.153 | 62 | 1771.418 | 1968.205 | 2164.878 | 2361.491 | 2597.471 |
| 64 | 1778.173 | 1975.648 | 2173.062 | 2370.52 | 2607.115 | 64 | 1778.563 | 1976.154 | 2173.659 | 2371.118 | 2608.103 |
| 66 | 1785.449 | 1983.765 | 2182.042 | 2380.347 | 2618.076 | 66 | 1785.709 | 1984.102 | 2182.439 | 2380.746 | 2618.736 |
| 68 | 1792.724 | 1991.883 | 2191.021 | 2390.173 | 2629.038 | 68 | 1792.854 | 1992.051 | 2191.22 | 2390.373 | 2629.368 |
| 70 | 1800. | 2000. | 2200. | 2400. | 2640. | 70 | 1800 | 2000 | 2200 | 2400 | 2640 |
| 72 | 1807.28 | 2008.136 | 2208.987 | 2409.831 | 2650.847 | 72 | 1807.146 | 2007.981 | 2208.787 | 2409.594 | 2650.571 |
| 74 | 1814.56 | 2016.272 | 2217.973 | 2419.662 | 2661.694 | 74 | 1814.293 | 2015.962 | 2217.574 | 2419.189 | 2661.142 |
| 76 | 1821.839 | 2024.409 | 2226.96 | 2429.494 | 2672.54 | 76 | 1821.439 | 2023.943 | 2226.361 | 2428.783 | 2671.712 |
| 78 | 1829.119 | 2032.545 | 2235.946 | 2439.325 | 2683.387 | 78 | 1828.586 | 2031.924 | 2235.148 | 2438.378 | 2682.283 |
| 80 | 1836.399 | 2040.681 | 2244.933 | 2449.156 | 2694.234 | 80 | 1835.732 | 2039.905 | 2243.935 | 2447.972 | 2692.854 |
| 82 | 1843.68 | 2048.777 | 2253.898 | 2458.979 | 2705.12 | 82 | 1842.87 | 2047.858 | 2252.703 | 2457.57 | 2703.479 |
| 84 | 1850.961 | 2056.873 | 2262.863 | 2468.802 | 2716.006 | 84 | 1850.008 | 2055.812 | 2261.471 | 2467.167 | 2714.104 |
| 86 | 1858.243 | 2064.969 | 2271.827 | 2478.626 | 2726.892 | 86 | 1857.145 | 2063.765 | 2270.24 | 2476.765 | 2724.728 |
| 88 | 1865.524 | 2073.065 | 2280.792 | 2488.449 | 2737.778 | 88 | 1864.283 | 2071.719 | 2279.008 | 2486.362 | 2735.353 |
| 90 | 1872.805 | 2081.161 | 2289.757 | 2498.272 | 2748.664 | 90 | 1871.421 | 2079.672 | 2287.776 | 2495.96 | 2745.978 |
| 92 | 1880.077 | 2089.298 | 2298.732 | 2508.108 | 2759.503 | 92 | 1878.575 | 2087.629 | 2296.563 | 2505.581 | 2756.556 |
| 94 | 1887.348 | 2097.435 | 2307.707 | 2517.944 | 2770.342 | 94 | 1885.73 | 2095.587 | 2305.35 | 2515.202 | 2767.134 |
| 96 | 1894.62 | 2105.572 | 2316.681 | 2527.78 | 2781.18 | 96 | 1892.884 | 2103.544 | 2314.138 | 2524.822 | 2777.711 |
| 98 | 1901.891 | 2113.709 | 2325.656 | 2537.616 | 2792.019 | 98 | 1900.039 | 2111.502 | 2322.925 | 2534.443 | 2788.289 |
| 100 | 1909.163 | 2121.846 | 2334.631 | 2547.452 | 2802.858 | 100 | 1907.193 | 2119.459 | 2331.712 | 2544.064 | 2798.867 |
| 102 | 1916.424 | 2129.946 | 2343.57 | 2557.242 | 2813.696 | 102 | 1914.331 | 2127.416 | 2340.481 | 2553.637 | 2809.459 |
| 104 | 1923.686 | 2138.05 | 2352.508 | 2567.032 | 2824.538 | 104 | 1921.468 | 2135.377 | 2349.251 | 2563.21 | 2820.05 |
| 106 | 1930.947 | 2146.151 | 2361.447 | 2576.822 | 2835.378 | 106 | 1928.606 | 2143.335 | 2358.02 | 2572.783 | 2830.642 |
| 108 | 1938.209 | 2154.253 | 2370.385 | 2586.612 | 2846.218 | 108 | 1935.743 | 2151.294 | 2366.79 | 2582.356 | 2841.233 |
| 110 | 1945.47 | 2162.355 | 2379.324 | 2596.402 | 2857.058 | 110 | 1942.881 | 2159.253 | 2375.559 | 2591.929 | 2851.825 |
| 112 | 1952.729 | 2170.464 | 2388.29 | 2606.225 | 2867.91 | 112 | 1950.019 | 2167.198 | 2384.322 | 2601.53 | 2862.393 |
| 114 | 1959.987 | 2178.574 | 2397.257 | 2616.047 | 2878.761 | 114 | 1957.156 | 2175.142 | 2393.085 | 2611.131 | 2872.961 |
| 116 | 1967.246 | 2186.683 | 2406.223 | 2625.87 | 2889.613 | 116 | 1964.294 | 2183.087 | 2401.847 | 2620.732 | 2883.53 |
| 118 | 1974.504 | 2194.793 | 2415.19 | 2635.692 | 2900.464 | 118 | 1971.431 | 2191.031 | 2410.61 | 2630.333 | 2894.098 |
| 120 | 1981.763 | 2202.902 | 2424.156 | 2645.515 | 2911.316 | 120 | 1978.569 | 2198.976 | 2419.373 | 2639.934 | 2904.666 |
| 122 | 1989.034 | 2214.637 | 2433.093 | 2655.312 | 2922.128 | 122 | 1985.699 | 2206.928 | 2428.137 | 2649.505 | 2915.249 |
| 124 | 1996.305 | 2226.373 | 2442.03 | 2665.109 | 2932.94 | 124 | 1992.829 | 2214.879 | 2436.901 | 2659.075 | 2925.832 |
| 126 | 2003.576 | 2238.108 | 2450.966 | 2674.906 | 2943.752 | 126 | 1999.958 | 2222.831 | 2445.666 | 2668.646 | 2936.415 |
| 128 | 2010.847 | 2249.844 | 2459.903 | 2684.703 | 2954.564 | 128 | 2007.088 | 2230.782 | 2454.43 | 2678.216 | 2946.998 |
| 130 | 2018.118 | 2261.579 | 2468.84 | 2694.5 | 2965.376 | 130 | 2014.218 | 2238.734 | 2463.194 | 2687.787 | 2957.581 |

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4.3 First article inspection. Prior to delivery of any mixed gas, a first cylinder shall be selected as specified in 4.4.3.2 and forwarded to the laboratory designated (see 6.2). The cylinder shall be subjected to the tests specified in 4.6.2, 4.6.4, 4.6.5, and tested for compliance with the minor constituent limitations specified in table I.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of:

(a) Individual inspection (4.4.2)
(b) Sampling inspection (4.4.3)

4.4.1 Place of inspection. Unless otherwise specified (see 6.2), quality conformance inspection on cylinders filled with mixed gas shall be conducted at the site of filling.

4.4.2 Individual inspection. Each cylinder filled with mixed gas shall be subjected to the leak and odor tests (see 4.6.1 and 4.6.2).

4.4.3 Sampling.

4.4.3.1 Lot definition. Each set of cylinders filled on the same manifold at the same time shall constitute a lot for the purposes of inspection and tests.

4.4.3.2 Sampling for inspection and tests. A sample of cylinders shall be selected at random from each lot in accordance with the procedures of MIL-STD-105 at inspection level S-3, as shown in table III.

Table III. Sampling for inspection and tests.

| Number of cylinders in lot | Number of cylinders in sample |
|---|---|
| 2 to 15 | 2 |
| 16 to 50 | 3 |
| 51 and over | 5 |

4.4.3.2.1 Each cylinder in the sample shall be subjected to the following tests:

| | |
|---|---|
| Fill pressure | (4.6.3) |
| Purity | (4.6.4) |
| Moisture | (4.6.5) |
| Examination of preparation for delivery | (4.6.6) |

4.4.3.2.2 The fill pressure test shall be performed prior to the purity and moisture tests. Each cylinder selected, after being subjected to the fill test, shall have approximately 1 cubic foot of gas withdrawn prior to withdrawing samples of gas for testing. If a sample cylinder is tested for moisture content in accordance with 4.6.5 or when any sample cylinder has had such an amount of gas withdrawn during testing that the internal pressure may be below the minimum acceptable value according to 4.6.3, the cylinder pressure shall be checked after completion of all tests. If the sample cylinder pressure is then found to be below the minimum acceptable value, the cylinder shall be filled to within the acceptable range, with the same mix ration as contained in cylinder.

4.4.3.3 Sampling, mix I and mix II. A sample of gas shall be taken from the mixed gas being manufactured to fulfill the contract and shipped to the laboratory specified (see 6.2) for testing to determine compliance with the requirements specified in table I. This sampling shall commence with initial production and, unless otherwise specified (see 6.2), shall be performed at a maximum interval of once every 45 calendar days for the duration of the contract. The sample of the mixed gas shall be forwarded in a sample bottle having a minimum volume of 276 cubic inches charged to 400 psig pressure. In addition, a comparable sample of the mixed gas shall be taken and analyzed within ten days of anticipated shipping date to ascertain that no selective leakage has occurred. Samples shall be tagged for identification (see 5.2.1).

4.4.4 Rejection.

4.4.4.1 Mixed gas cylinders. If any cylinder in the sample fails to meet all the requirements of this specification, the lot shall be rejected. Each cylinder of a rejected lot shall be subjected to the tests specified in 4.4.3.2. If the cause of failure cannot be located or corrected in sufficient time to avoid delays in completion of the contract or order, the procuring activity shall be immediately notified. If the cause of failure is expeditiously located and corrected, each cylinder in the rejected lot which was found to be defective shall be processed as follows, based on the test failed.

4.4.4.2 Leak test. Cylinders found to leak through the valve or at the junction between the valve and the cylinder shall be emptied to zero psig pressure, and the cylinder valve repaired or replaced in accordance



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with 3.4.2.4 and 3.4.2.5. The cylinder shall then be evacuated in accordance with 3.4.2.8 before refilling.

4.4.4.3 Odor test. When cylinders charged from the same manifold at the same time fail to pass the odor test of 4.6.2, all the cylinders shall be emptied to zero psig pressure and evacuated as specified in 3.4.2.8 before refilling. When one cylinder or a portion of the gas of one cylinder fails to pass the odor test, the cylinder shall be considered unsatisfactory for refill and shall be tagged as specified in 3.4.4.

4.4.4.4 Filling pressure test. A cylinder having more than the maximum allowable filling pressure shall be emptied to within the acceptable range and subjected to the remainder of the required tests as though it had satisfactorily completed the pressure filling test. A cylinder having less than the specified minimum filling pressure shall be charged to within the acceptable pressure range and retested as a freshly filled cylinder. However, unless the cylinder is filled alone, it shall not be selected as a sample from the new lot of filled cylinders.

4.4.4.5 Purity and moisture tests. Cylinders failing to pass the purity or moisture tests (see 4.6.4 and 4.6.5) shall be emptied to zero psig and evacuated as specified in 3.4.2.8.

4.5 Test conditions.

4.5.1 Chemicals. Chemical tests shall be made with analytical reagent grade chemicals and distilled water.

4.5.2 Temperatures. Temperatures shall be measured using thermocouples or thermometers. Thermocouples shall be connected to accurately calibrated potentiometers. Thermometers shall be of laboratory accuracy and shall have a sufficient number of scale divisions to be read within 1/2°F.

4.5.3 Pressures. Pressures shall be measured by means of accurately calibrated Bourdon tube gages or any other laboratory-type pressure measuring device capable of being accurately read within 15 psig. For the purposes of this specification, a gage or other laboratory-type pressure measuring device having scale increments of 10 psig or smaller is considered satisfactory for reading within 15 psig.

4.5.4 Tests on mixed gas cylinders shall not permit water or other liquid to be brought into contact with the valve outlet. This includes the test of 4.6.1.

4.6 Inspection methods.

4.6.1 Leak test. Each cylinder shall be tested for leakage by brushing a soap solution over the cylinder and all portions of the valve, except the valve outlet. Each cylinder shall be tested for leakage through the valve when closed, by means of a tube from the valve outlet to a container of liquid. Bubbling of gas through the liquid shall be evidence of leaking through the valve and cause for rejection. Care shall be taken to insure that water or other liquid is not brought into contact with the valve outlet.

4.6.2 Odor tests. Mixed gas shall be tested for odor by cracking open the cylinder valve and smelling the escaping gas. Mixed gas is odorless and tasteless. Filled cylinders having an odor present in the gas shall be rejected.

4.6.3 Filling pressure test. A cylinder shall be tested for proper filling pressure by attaching a gage of the type described in 4.5.3 to the valve outlet and by attaching either a thermocouple or thermometer of the type described in 4.5.2 to the cylinder wall. If a thermometer is used, the bulb shall be protected from alien temperatures by a fireproof insulating material secured by either spring clips or tape. The insulation shall not be applied between the bulb and the cylinder wall. The cylinder shall stand until it stabilizes at approximately room temperature. The cylinder valve shall be opened and the internal pressure observed on the gage. If the internal pressure differs from the applicable pressure value shown in table II corresponding to the stabilized cylinder temperature by more than 25 psig, the sample cylinder and all other cylinders filled from the same manifold at the same time shall be rejected.

4.6.4 Purity. Gas shall be tested for purity in accordance with (a) a method of U. S. P., or (b) any other method approved by the procuring activity.

4.6.5 Moisture tests. Gas shall be tested for water vapor content with an instrument approved by the procuring activity. Samples indicating a water vapor content of more than 0.005 milligram per liter of gas at 70°F. and 760 millimeters of mercury pressure shall constitute cause for rejection of the lot.

4.6.6 Examination of preparation for delivery. Mixed gas cylinders shall be examined to insure that preparation for delivery conforms to the applicable requirements of section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)



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5.1 Packaging and packing. The packaging and packing for all levels shall be as specified in 5.1.1.

5.1.1 Levels A, B, or C. All cylinders shall be maintained, packed and marked in accordance with current Department of Transportation Regulations (D. O. T.), Title 49 (CFR 171 to 190).

5.1.2 Palletised loads. When specified (see 6.2), the cylinders shall be palletised in accordance with MIL-STD-147, except that the unit quantity on the pallet shall not exceed 2100 pounds.

5.2 Marking. In addition to any special marking required (see 6.2), and herein, all cylinders shall be marked in accordance with MIL-STD-129 and D. O. T. regulation title 49 CFR 171-190. A tag bearing the following information shall be firmly wired to the cylinder valve: Mixed gas, mix I or mix II, filling date, stock number, quantity contained, name of contractor, lot number, and Government order or contract number. The requirements of MIL-STD-101 shall apply for Government-owned cylinders. Cylinders shall be marked on the necked-in portion as follows:

| Mixed gas diver's breathing for military purposes only | Helium-Oxygen Mix I | Helium-Oxygen Mix II |
|---|---|---|

5.2.1 The sample submitted to the laboratory designated by the procuring activity shall be marked with information indicating where and when the sample was obtained and the lot number.

6. NOTES

6.1 Intended use. Mixed gas is intended for use in diving operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of mix required (see 1.2).
(c) Whether cylinders will be furnished by the contractor without cost to the Government for return when empty, whether the Government will furnish the cylinders for filling, or whether cylinders will be furnished by the contractor and remain the property of the Government (see 3.4.1).
(d) If the contractor is to be responsible for hydrostatic testing of cylinders and detailed procedure for testing (see 3.4.2.2).
(e) Instructions for disposition of cylinder containers found unsatisfactory for filling (see 3.4.4).
(f) Nominal filling pressure required for cylinders (see 3.5.1).
(g) Points of inspection if other than specified in 4.4.1 (see 4.3 and 4.4.3.3).
(h) When palletisation is required (see 5.1.2).
(i) Special marking (see 5.2).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when materials and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 The basis of purchase should be the conditions. Standard cubic foot (1 atom, 70°F.).

6.5 First article inspection.

6.5.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 6830-N067)



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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Naval Ship Engineering Center
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Prince George's Center
Hyattsville, Maryland 20782

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